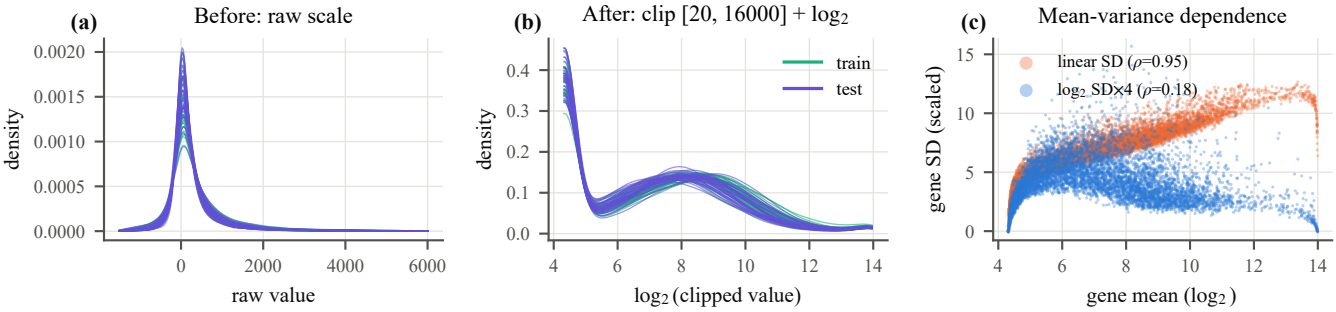

(a)
Before: raw scale
density
0.0020
0.0015
0.0010
0.0005
0.0000
0
2000
4000
6000
raw value
(b)
After: clip [20, 16000] + log2
density
0.4
0.3
0.2
0.1
0.0
train
test
4
6
8
10
12
14
log2 (clipped value)
(c)
Mean-variance dependence
gene SD (scaled)
15
10
5
0
linear SD (ρ=0.95)
log2 SD×4 (ρ=0.18)
4
6
8
10
12
14
gene mean (log2 )